

Cross-Sectional and Time-Series Asset Pricing Tests

Testing the CAPM — Roll's critique, Fama–MacBeth,
Fama–French, and Gibbons–Ross–Shanken

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MGT 595 · Quantitative Investing

Lecture 2

What this deck covers

- **Testable implications of the CAPM**

Sharpe–Lintner vs Black, SML vs CML

- **Roll's critique**

Testability, market proxies, sensitivity

- **Cross-sectional tests: Fama–MacBeth (1973)**

Two-pass regressions, portfolio interpretation, errors-in-variables

- **Practical data issues**

Microstructure, regression phenomenon, beta estimation

- **Fama–French (1992): is beta dead?**

Size, book-to-market, leverage, E/P

- **Time-series tests: Gibbons–Ross–Shanken (1989)**

Multivariate F-test, intercepts as alphas, geometry

- **Multifactor models: Fama–French (1993)**

SMB, HML, TERM, DEF

- **What rejecting the model buys us**

Testable implications of the CAPM

Recap: the CAPM equation

Two statements of one model: with r_f , and without.

Sharpe–Lintner CAPM

Free borrowing/lending at r_f :

$$\mathbb{E}[\tilde{r}_i] = r_f + \beta_i (\mathbb{E}[\tilde{R}_M] - r_f)$$
$$\underbrace{\mathbb{E}[\tilde{r}_i] - r_f}_{\text{risk premium}} = \underbrace{\beta_i}_{\text{risk}} \times \underbrace{(\mathbb{E}[\tilde{R}_M] - r_f)}_{\text{price of risk}}$$

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Black (zero-beta) CAPM

No r_f , or constrained borrowing/lending:

$$\mathbb{E}[\tilde{r}_i] = \mathbb{E}[\tilde{R}_z] + \beta_i (\mathbb{E}[\tilde{R}_M] - \mathbb{E}[\tilde{R}_z])$$

$\tilde{R}_z$: return on an efficient portfolio uncorrelated with M (the zero-beta portfolio).

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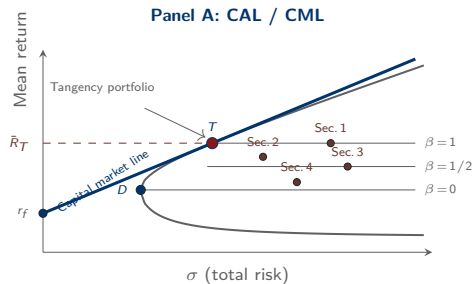
The E - β relation to be tested

$$\mathbb{E}[\tilde{r}_i] = \gamma_1 + \gamma_2 \beta_i, \quad \gamma_1 = r_f \text{ (or } \mathbb{E}[\tilde{R}_z]), \quad \gamma_2 = \mathbb{E}[\tilde{R}_M] - \gamma_1$$

Expected return is linear in β — and β is the only thing that matters.

The CML and the SML

Same equilibrium, two diagrams. SML is the more universal one.

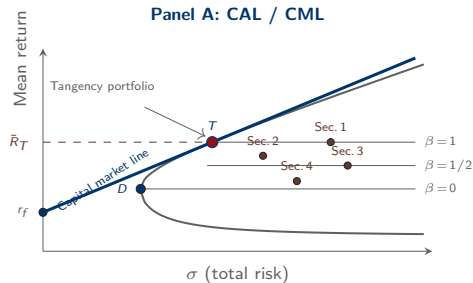


Total risk = systematic + idiosyncratic
= non-diversifiable + diversifiable

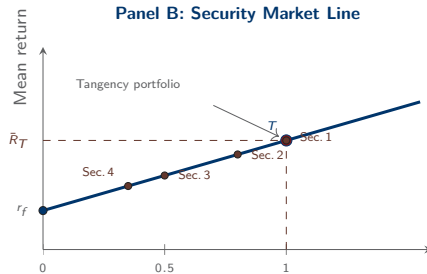
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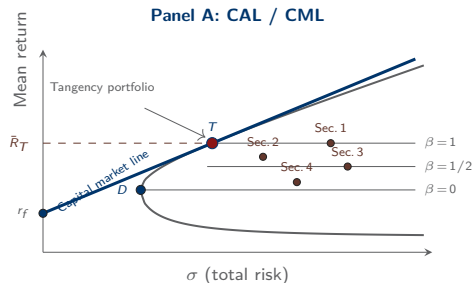
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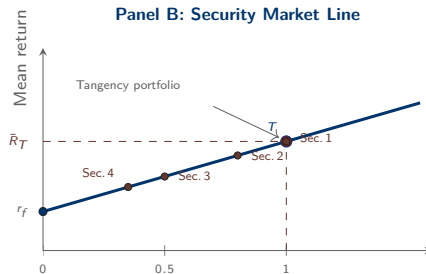
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β with return of the tangency portfolio
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Key distinction

All portfolios — efficient or not — must lie on the SML. Only the *efficient* ones lie on the CML. Two assets with the same $\mathbb{E}[r]$ may have different σ , but they must share the same β .

Total vs systematic risk: two diagrams

CML axis: total risk. SML axis: the priced part only.

The CML axis: total risk σ

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_M^2}_{\text{systematic}} + \underbrace{\sigma^2(\varepsilon_i)}_{\text{idiosyncratic}}$$

Systematic: non-diversifiable

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The SML axis: β alone

$$\beta_i = \frac{\text{Cov}(\tilde{r}_i, \tilde{R}_M)}{\text{Var}(\tilde{R}_M)}$$

Captures only the systematic piece

In equilibrium: no compensation for diversifiable risk

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In equilibrium: no compensation for diversifiable risk

Same $\mathbb{E}[r]$, different σ^2

Investments with the same $\mathbb{E}[r]$ can have different total variances, but must share the same β . What prices: β (covariance), not variance.

Intuition behind the CAPM

Why high- β stocks are risky, and why variance alone is not.

High- β stocks are risky because. . .

. . . they pay off when you need the money *least* — i.e. when the overall market is doing well

For anyone to hold such a security, it must offer a **high expected return**

Conversely, low- β assets serve as hedges and command *lower* expected returns

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Compensation is for covariance, not variance

An asset with huge variance but zero covariance with the market:

CAPM says $\mathbb{E}[\tilde{r}_i] = r_f$ — no premium

Hold it alone: trouble

Hold it in a diversified portfolio: variance washes out, no compensation needed

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Bottom line

Pricing rewards *co-movement with bad times*, not raw volatility.

Four testable implications

$$\mathbb{E}[\tilde{r}_i] = r_f + \beta_i(\mathbb{E}[\tilde{R}_M] - r_f)$$

1. $\mathbb{E}[\tilde{r}_i]$ and β_i should be **linearly related**.
2. β is the **only** measure of risk needed to explain differences in expected returns.
3. M must lie on the positively-sloped segment of the frontier: $\mathbb{E}[\tilde{R}_M] > \mathbb{E}[\tilde{R}_z]$.
4. In the standard model with a risk-free asset, $\mathbb{E}[\tilde{R}_z] = r_f$.

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Sharpe–Lintner version: in excess returns

Estimate by running the time-series regression

$$r_{it} - r_{ft} = \alpha_i + \beta_i(R_{Mt} - r_{ft}) + \varepsilon_{it}$$

- (1) The intercept α_i is **zero** ← **time-series test**
- (2) β completely captures the cross-sectional variation in expected excess returns ← **cross-sectional test**
- (3) The market risk premium is positive, $\mathbb{E}[\tilde{R}_m] - r_f > 0$

Testing the Black (zero-beta) version

When there is no risk-free asset, the regression is run in raw returns.

Estimate

$$r_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$$

1. The intercept satisfies $\alpha_i = \mathbb{E}[\tilde{R}_z](1 - \beta_i)$ for all stocks i .
2. β completely captures the cross-sectional variation of expected returns.
3. The market premium is positive: $\mathbb{E}[\tilde{R}_m] - \mathbb{E}[\tilde{R}_z] > 0$.

Roll's critique

Roll's critique: one testable hypothesis, not many

Only one implication of the CAPM is genuinely testable.

The single hypothesis

The market portfolio M is mean–variance efficient.

All other implications — including the linear $\mathbb{E}[r]-\beta$ relation — **follow from** the mean–variance efficiency of M and are not independently testable.

In any sample, there always exist infinitely many ex-post efficient portfolios. For each one, the linear relation in betas computed against *that* portfolio holds in-sample by construction — regardless of whether the *true* market is mean–variance efficient.

⇒ The CAPM is not testable unless you can measure the *true* market portfolio of *all* assets.

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The proxy problem

Using a market proxy creates three issues simultaneously:

1. The proxy can be efficient when the true market is not.
2. The proxy can be inefficient when the true market is.
3. Therefore, on rejection, we never know if it's the model or the proxy.

How sensitive are inferences to the proxy?

Empirically: more bark than bite. But not entirely.

Stambaugh (1982)

Tests use various proxies that add real estate and bonds to equities

Inferences are **similar** across proxies

Suggests Roll's critique is **not** an important empirical issue

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Find that it **helps** explain returns

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Estimate the maximum correlation between proxy and true market needed to overturn rejection of the CAPM

If $\text{Corr}(\text{proxy}, \text{true}) > 0.70$, then rejection using the proxy implies rejection using the true market

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Practical implication

As long as we believe the correlation between our proxy and the true market is high, our tests are valid.

A note on time: the CAPM is static

No time dimension in the CAPM, but the tests still need iid.

The maintained assumption

For empirical work, we assume returns are jointly multivariate normal and **iid through time**.

Strong assumption, but theoretically consistent with the single-period CAPM

Implies the CAPM holds period-by-period

Makes time-series averages valid estimators of expectations

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Why we still care

Despite its strong assumptions and the empirical rejections to come, the CAPM (and the betas it produces) remain useful for:

Cost of capital

Performance evaluation (Jensen's α , Sharpe ratio)

Event studies and abnormal returns

**Cross-sectional
tests:
Fama–MacBeth
(1973)**

The Fama–MacBeth (1973) cross-sectional regression

Run the cross-section, period by period, and average the slopes.

Cross-sectional regression at each date t

$$\tilde{r}_{it} = \tilde{\gamma}_{1t} + \tilde{\gamma}_{2t} \hat{\beta}_i + \tilde{\eta}_{it}, \quad i = 1, \dots, N$$

Estimate the regression *across all stocks* at each t .

Average $\hat{\gamma}_{1t}$ over time: compare to r_f .

Average $\hat{\gamma}_{2t}$ over time: compare to $\mathbb{E}[\tilde{R}_M] - \mathbb{E}[\tilde{R}_z]$.

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An augmented regression: testing alternatives

$$\tilde{r}_{it} = \tilde{\gamma}_{1t} + \tilde{\gamma}_{2t} \hat{\beta}_i + \tilde{\gamma}_{3t} \hat{\beta}_i^2 + \tilde{\gamma}_{4t} \hat{\sigma}(\varepsilon_i) + \tilde{\eta}_{it}$$

Under the CAPM null: $\mathbb{E}[\tilde{\gamma}_{3t}] = 0$ (linearity) and $\mathbb{E}[\tilde{\gamma}_{4t}] = 0$ (no premium for idiosyncratic risk)

Captures both the null and economically meaningful alternatives

Step 1: time-series regression for each asset

Get the right-hand-side variables ready before doing any cross-section.

For each asset $i = 1, \dots, N$, run the time-series regression

$$\underbrace{r_{it} - r_{ft}}_{(T \times 1)} = \alpha_i + \hat{\beta}_i \underbrace{(R_{mt} - r_{ft})}_{(T \times 1)} + \underbrace{\varepsilon_{it}}_{(T \times 1)}$$

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What we extract

For each asset i we save

$$\hat{\beta}_i, \quad \hat{\sigma}^2(\varepsilon_i).$$

Stack betas into the $(N \times 1)$ vector

$$\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_N)'.$$

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Why this comes first

β is unobservable — it must be *estimated* from the data before it can appear as a regressor.

$\hat{\sigma}^2(\varepsilon_i)$ will play the role of the idiosyncratic-risk variable in the augmented regression.

This generates errors-in-variables — handled later.

Step 2: the cross-section and the null hypotheses

Cross-sectional regression in vector form

$$\underbrace{\bar{\mathbf{r}}}_{(N \times 1)} = \gamma_1 \mathbf{1} + \gamma_2 \underbrace{\boldsymbol{\beta}}_{(N \times 1)} + \gamma_3 \underbrace{\boldsymbol{\beta}^2}_{(N \times 1)} + \gamma_4 \underbrace{\boldsymbol{\sigma}_\varepsilon^2}_{(N \times 1)} + \tilde{\boldsymbol{\eta}}$$

where $\bar{\mathbf{r}} = (\bar{r}_1, \dots, \bar{r}_N)'$ is the vector of average returns across assets.

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Implied expected return under the CAPM

$$\mathbb{E}[\tilde{r}_i] = \gamma_1 + \gamma_2 \beta_i + \gamma_3 \beta_i^2 + \gamma_4 \sigma_{\varepsilon_i}^2$$

The Sharpe–Lintner CAPM predicts

$$\mathbb{E}[\tilde{r}_i] = r_f + (\mathbb{E}[\tilde{R}_M] - r_f) \beta_i + 0 \cdot \beta_i^2 + 0 \cdot \sigma_{\varepsilon_i}^2$$

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The four null hypotheses

Read off the predicted slopes:

$$H_0 : \mathbb{E}[\gamma_1] = r_f$$

$$\mathbb{E}[\gamma_2] = \mathbb{E}[\tilde{R}_M] - r_f > 0$$

$$\mathbb{E}[\gamma_3] = 0$$

$$\mathbb{E}[\gamma_4] = 0$$

Test each by time-averaging the monthly $\hat{\gamma}$ s and forming a t -statistic.

Portfolio interpretation: each γ as a portfolio return

Take a portfolio p with weights w_i , expand $\mathbb{E}[R_p]$, and read off four conditions.

$$\mathbb{E}[R_p] = \sum_{i=1}^N w_i \mathbb{E}[r_i] =$$

=

$$\begin{aligned} &\sum_i w_i \\ &\sum_i w_i \beta_i \\ &\sum_i w_i \beta_i^2 \\ &\sum_i w_i \sigma_{\varepsilon_i}^2 \end{aligned}$$

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$$\begin{aligned}\mathbb{E}[R_p] &= \sum_{i=1}^N w_i \mathbb{E}[r_i] = \sum_{i=1}^N w_i \left(r_f + (\mathbb{E}[\tilde{R}_M] - r_f) \beta_i + \gamma_3 \beta_i^2 + \gamma_4 \sigma_{\varepsilon_i}^2 \right) \\ &= r_f \underbrace{\sum_i w_i}_{\boxed{1}} + (\mathbb{E}[\tilde{R}_M] - r_f) \underbrace{\sum_i w_i \beta_i}_{\boxed{2}} + \gamma_3 \underbrace{\sum_i w_i \beta_i^2}_{\boxed{3}} + \gamma_4 \underbrace{\sum_i w_i \sigma_{\varepsilon_i}^2}_{\boxed{4}}\end{aligned}$$

1: $\mathbb{E}[R_p] = r_f$	
$\sum_i w_i$	1
$\sum_i w_i \beta_i$	0
$\sum_i w_i \beta_i^2$	0
$\sum_i w_i \sigma_{\varepsilon_i}^2$	0

Pick the weights w so that one constraint = 1 and the others = 0. The resulting portfolio return equals the corresponding γ (up to r_f).

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	1: $\mathbb{E}[R_p] = r_f$	2: $\mathbb{E}[R_p] = \mathbb{E}[\tilde{R}_M] - r_f$
$\sum_i w_i$	1	0
$\sum_i w_i \beta_i$	0	1
$\sum_i w_i \beta_i^2$	0	0
$\sum_i w_i \sigma_{\varepsilon_i}^2$	0	0

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$\sum_i w_i$	1	0	0
$\sum_i w_i \beta_i$	0	1	0
$\sum_i w_i \beta_i^2$	0	0	1
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Each γ is the return on a minimum-variance characteristic portfolio

Minimum-variance portfolios subject to the corresponding combination of constraints.

$\hat{\gamma}_1$: zero- β

$\min_w w' V w \quad \text{s.t.}$

$$w' \mathbf{1} = 1$$

$$w' \beta = 0$$

$$w' \beta^2 = 0$$

$$w' \sigma_\varepsilon^2 = 0$$

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$\hat{\gamma}_4$: σ_ε^2 -loaded

$\min_w w'Vw \text{ s.t.}$

$$\begin{aligned}w'1 &= 0 \\w'\beta &= 0 \\w'\beta^2 &= 0 \\w'\sigma_\varepsilon^2 &= 1\end{aligned}$$

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Minimum-variance portfolios subject to the corresponding combination of constraints.

$\hat{\gamma}_1$: zero- β	$\hat{\gamma}_2$: unit- β	$\hat{\gamma}_3$: β^2 -loaded	$\hat{\gamma}_4$: σ_ε^2 -loaded
$\min_w w'Vw$ s.t.	$\min_w w'Vw$ s.t.	$\min_w w'Vw$ s.t.	$\min_w w'Vw$ s.t.
$w'\mathbf{1} = 1$	$w'\mathbf{1} = 0$	$w'\mathbf{1} = 0$	$w'\mathbf{1} = 0$
$w'\beta = 0$	$w'\beta = 1$	$w'\beta = 0$	$w'\beta = 0$
$w'\beta^2 = 0$	$w'\beta^2 = 0$	$w'\beta^2 = 1$	$w'\beta^2 = 0$
$w'\sigma_\varepsilon^2 = 0$	$w'\sigma_\varepsilon^2 = 0$	$w'\sigma_\varepsilon^2 = 0$	$w'\sigma_\varepsilon^2 = 1$

Putting them together

The cross-sectional fitted values are $\hat{\mathbf{r}} = \hat{\gamma}_1\mathbf{1} + \hat{\gamma}_2\beta + \hat{\gamma}_3\beta^2 + \hat{\gamma}_4\sigma_\varepsilon^2$ — and each $\hat{\gamma}_k$ is the return on the corresponding zero-cost (or unit-cost), characteristic-mimicking, minimum-variance portfolio.

The time series of cross-sectional regressions

Run one cross-section per month; collect the slopes through time; average.

The monthly regressions, $t = 1, 2, \dots, T$

$$\begin{aligned}t = 1 : \quad \widetilde{\mathbf{r}}_1 &= \gamma_{1,1}\mathbf{1} + \gamma_{2,1}\widehat{\boldsymbol{\beta}} + \widetilde{\boldsymbol{\eta}}_1 \\t = 2 : \quad \widetilde{\mathbf{r}}_2 &= \gamma_{1,2}\mathbf{1} + \gamma_{2,2}\widehat{\boldsymbol{\beta}} + \widetilde{\boldsymbol{\eta}}_2 \\&\vdots \\t = T : \quad \widetilde{\mathbf{r}}_T &= \gamma_{1,T}\mathbf{1} + \gamma_{2,T}\widehat{\boldsymbol{\beta}} + \widetilde{\boldsymbol{\eta}}_T\end{aligned}$$

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Fama–MacBeth point estimate and standard error

$$\hat{\gamma}_2 = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{2,t}, \quad \text{s.e.}(\hat{\gamma}_2) = \frac{1}{\sqrt{T-1}} \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{\gamma}_{2,t} - \hat{\gamma}_2)^2}$$

Why average the slopes? Fama–MacBeth vs. one big regression

Two ways to use the same data: point estimates agree, standard errors do not.

Alternative: averages on averages

Run **one** cross-sectional regression of time-averaged returns on betas:

$$\bar{r}_i = \gamma_1 + \gamma_2 \hat{\beta}_i + \bar{\eta}_i, \quad \bar{r}_i = \frac{1}{T} \sum_t \tilde{r}_{it}$$

With the *same* regressor $\hat{\beta}_i$ every month, the OLS slope here is **numerically identical** to the FM time-average $\hat{\gamma}_2$.

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OLS standard errors treat the N residuals $\bar{\eta}_i$ as **independent** across assets. They are not: returns load on *common shocks* (market, industry, sector), so the $\bar{\eta}_i$ are strongly **cross-sectionally correlated**.

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Consequence: misspecified standard errors and t -stats

Ignoring the cross-correlation makes the single regression behave as if it had N independent observations when the *effective* sample is far smaller. The OLS s.e. on $\hat{\gamma}_2$ is **biased downward** — often by a large factor — so t -statistics are **overstated** and the CAPM can be “confirmed” (or rejected) spuriously. Averaging the data first also throws away the time-series information needed to estimate the true sampling variation of the slope.

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The Fama–MacBeth fix: treat each month’s slope $\hat{\gamma}_{2,t}$ as one draw. Cross-sectional correlation in $\tilde{\eta}_t$ is *absorbed* into the month-by-month variation of the slopes, and monthly slopes are close to serially **independent** — so the time-series s.e. on the previous slide is valid. Remaining caveats: it still assumes the $\hat{\gamma}_{2,t}$ are uncorrelated over time (autocorrelation would bias the s.e.), and it inherits the errors-in-variables problem from estimated betas — next.

Three classes of problems can mislead the regressions.

Microstructure pitfalls

Non-synchronous trading: not all stocks trade every minute; biases $\hat{\beta}$ for thin stocks

Bid-ask bounce: spurious negative autocorrelation in returns

Price discreteness: tick sizes induce noise

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$\hat{\beta}_i = \beta_i + \tilde{e}_i$ is measured with noise.

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Derived on the next slide.

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Regression phenomenon

Extreme $\hat{\beta}$ s revert toward 1 in subsequent periods

Sorting on noisy estimates picks up estimation error, not true beta variation

Fix: sort first, average within portfolios

The errors-in-variables (EIV) problem: in detail

A noisy regressor pushes the cross-sectional slope toward zero.

Setup

At date t , the cross-section is run with the *estimated* betas:

$$\tilde{\mathbf{r}}_t = \gamma_{1,t}\mathbf{1} + \gamma_{2,t}\hat{\boldsymbol{\beta}} + \tilde{\boldsymbol{\eta}}_t, \quad \hat{\beta}_i = \beta_i + \tilde{\mathbf{e}}_i, \quad \mathbb{E}[\hat{\beta}_i] = \beta_i.$$

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Cross-sectional slope: bias derivation

$$\hat{\gamma}_2 = \frac{\text{Cov}(r, \hat{\beta})}{\text{Var}(\hat{\beta})}$$

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Consequence: attenuation bias

As $\text{Var}(\tilde{e}) \uparrow$, $\hat{\gamma}_2 \rightarrow 0$. The noisier the beta estimate, the more the cross-sectional slope is biased toward zero — exactly the wrong direction for a power test of the CAPM.

Mitigating EIV: form portfolios first

Diversification kills idiosyncratic noise in the beta estimate.

Sampling distribution of $\hat{\beta}_i$

Under the time-series regression assumptions,

$$\hat{\beta}_i \sim \mathcal{N}\left(\beta_i, \frac{\sigma^2(\varepsilon_i)}{\sum_{t=1}^T (R_{m,t} - \bar{r}_{f,t})^2}\right) \approx \mathcal{N}\left(\beta_i, \frac{\sigma^2(\varepsilon_i)}{T \sigma_M^2}\right)$$

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Portfolios reduce residual variance

For a portfolio $R_p = \sum_{i=1}^N w_i r_i$:

$$\varepsilon_{p,t} = \sum_{i=1}^N w_i \varepsilon_{i,t} \implies \overline{\sigma^2(\varepsilon_p)} < \overline{\sigma^2(\varepsilon_i)}$$

Then run the time-series regression

$$R_{p,t} = \alpha_p + \beta_p (R_{m,t} - r_{f,t}) + \varepsilon_{p,t}$$

$\Rightarrow \hat{\beta}_p$ has smaller standard error than any individual $\hat{\beta}_i$.

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The trade-off

Forming portfolios reduces $\text{Var}(\tilde{e})$ — but it also reduces $\text{Var}(\beta)$, the cross-sectional spread we need to identify γ_2 .

$\implies$ FF construction: **sort to maximize β -spread, then average within portfolios.**

**Fama–French
(1992): is beta
dead?**

Fama–French (1992): the indictment

Two Sharpe–Lintner–Black implications, on trial.

What FF92 test

1. Expected returns are a positive linear function of market β .
2. Market β **suffices** to describe the cross-section of expected returns.

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Size (Banz 1981): small stocks earn “too much” for their β , large stocks “too little”

Leverage (Bhandari 1988): matters above and beyond β and size

Book-to-market, BE/ME (Stattman 1980; Rosenberg, Reid & Lanstein 1985): higher BE/ME, higher returns

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Ball (1978): all four are scaled-price ratios — each puts lagged price in the denominator.

The headline findings of FF92

Main results: “beta is dead”

1. Average stock returns are **not** positively related to market β .
2. β does **not** capture the cross-sectional variation in average returns.
3. The combination of **size and BE/ME** captures a sizeable portion of cross-sectional variation *and* wipes out the explanatory power of leverage and E/P.

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What FF92 do that is new

Examine the *joint* significance and *interaction* of β , size, BE/ME, leverage, and E/P

Carefully examine the role of β in the cross-section

Use **independent variation** in β and size by double-sorting — the key methodological move

FF92 data and sample construction

A careful design: several deliberate choices.

Sample

Universe: All non-financial NYSE/AMEX/NASDAQ stocks

Source: CRSP returns merged with Compustat accounting

Period: 1962–1989

Financials excluded because their leverage has a different meaning

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Accounting data from fiscal year $t - 1$ are matched with returns starting in **July of year t** — ensuring the data were available to investors before the returns being explained.

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BE/ME, E/P: use ME at end of December $t - 1$ in the denominator

Size (ln ME): use ME at end of **June** of year t

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Selection biases

Stock must have price for June t and Dec $t - 1$

Must have ≥ 24 of past 60 months of return history

These requirements may bias toward survivors

Estimating betas without confounding β and size

A two-way sort separates the two effects: the heart of FF92.

1. In June of year t , sort stocks into **10 size deciles** using NYSE breakpoints.

Why NYSE breakpoints? To avoid swamping the small-size deciles with thousands of tiny NASDAQ stocks.

2. Within each size decile, sort into **10 portfolios on pre-ranking β** .

Pre-ranking β estimated using last 24–60 months ending June t .

Generates $10 \times 10 = \mathbf{100}$ size- β portfolios.

3. Compute equal-weighted monthly returns July t to June $t + 1$. Repeat every year.

4. Estimate **full-period post-ranking β s** using the entire 1963–1990 sample, allowing for non-synchronous trading:

$$R_{it} = \alpha_i + \beta_i R_{Mt} + \delta_i R_{M,t-1} + \eta_{it}, \quad \hat{\beta}_i^{\text{post}} = \hat{\beta}_i + \hat{\delta}_i$$

5. Assign each stock the post-ranking β of its current portfolio. Stocks move across portfolios as their characteristics change.

FF92 Table I: size and β within size deciles

Within size deciles β varies independently; returns barely move.

	All	Low- β	β -3	β -5	β -7	β -9	High- β
<i>Panel A: Average monthly returns (%)</i>							
All	1.25	1.34	1.36	1.33	1.24	1.25	1.14
Small-ME	1.52	1.71	1.79	1.50	1.37	1.50	1.42
ME-5	1.29	1.34	1.39	1.42	1.13	1.18	1.08
Large-ME	0.89	1.01	1.10	0.93	1.03	0.74	0.56
<i>Panel B: Post-ranking βs</i>							
All	—	0.87	1.09	1.26	1.35	1.52	1.72
Small-ME	1.44	1.05	1.28	1.40	1.49	1.64	1.79
Large-ME	0.92	0.57	0.78	0.95	1.02	1.11	1.32

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What the table says

Read **down**: large size $\Rightarrow$ lower returns (size effect, strong)

Read **across** (within a size decile): β varies a lot, but returns hardly move

Sorting independently on β within size produces **no relation** between β and average returns

FF92 Table III: Fama–MacBeth regressions

Headline: $\ln(BE/ME)$ and $\ln(ME)$ crush β .

β	$\ln(ME)$	$\ln(BE/ME)$	$\ln(A/ME)$	$\ln(A/BE)$	$E(+)/P$
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Reading the table

Avg slope (t-stat). 7/63–12/90, $\sim 2,267$ stocks/mo.

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−0.37	−0.17				
(−1.21)	(−3.41)				

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		0.50 (5.71)			

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−0.37 (−1.21)	−0.17 (−3.41)				
		0.50 (5.71)			
	−0.11 (−1.99)	0.35 (4.44)			

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Size kills E/P; BE/ME kills leverage

FF92 conclusions and the open question

What FF92 establish

Allowing for variation in β that is unrelated to size, **there is no relation between β and average returns.**

The opposite roles of market and book leverage in returns are captured by $\ln(\text{BE}/\text{ME})$.

The power of E/P is captured by size and BE/ME .

The size and BE/ME effects are robust across subperiods.

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Rational story

Size and BE/ME proxy for **state variables** or systematic risks not captured by the market portfolio — a multifactor model.

But why these variables?

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Behavioral story

Size and BE/ME reflect **mispricing** — investor overreaction, extrapolation, sentiment — not compensation for risk.

Time-series tests: Gibbons–Ross– Shanken (1989)

Time-series tests: N regressions in parallel

If the CAPM is right, all the alphas should be zero, jointly.

Estimate N time-series regressions (one per asset)

$$\begin{aligned}\underbrace{r_{1,t} - r_{f,t}}_{(T \times 1)} &= \hat{\alpha}_1 + \hat{\beta}_1 \underbrace{(R_{m,t} - r_{f,t})}_{(T \times 1)} + \underbrace{\varepsilon_{1,t}}_{(T \times 1)} \\ &\vdots \\ \underbrace{r_{N,t} - r_{f,t}}_{(T \times 1)} &= \hat{\alpha}_N + \hat{\beta}_N \underbrace{(R_{m,t} - r_{f,t})}_{(T \times 1)} + \underbrace{\varepsilon_{N,t}}_{(T \times 1)}\end{aligned}$$

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The joint null hypothesis

$$H_0 : \hat{\alpha}_1 = \hat{\alpha}_2 = \dots = \hat{\alpha}_N = 0$$

Stack into the $(N \times 1)$ vector $\hat{\alpha} = (\hat{\alpha}_1, \dots, \hat{\alpha}_N)'$.

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Why a joint test, not N separate t -tests

Individual t -tests on each $\hat{\alpha}_i$ ignore the cross-equation correlation in $\varepsilon_{i,t}$, inflate Type-I error, and miss joint deviations. The GRS test uses the full residual covariance matrix $\hat{\Sigma}$.

Sampling distribution of $\hat{\alpha}$

Under iid multivariate normal returns, the alpha vector has a tractable distribution.

Distribution of the stacked intercept estimator

$$\hat{\alpha} \sim \mathcal{N}\left(\alpha, \frac{1}{T} \left[1 + \frac{\hat{\mu}_m^2}{\hat{\sigma}_m^2}\right] \hat{\Sigma}\right) \quad (N \times 1)$$

Note the factor $[1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2]$: the squared Sharpe ratio of the market enters the variance of $\hat{\alpha}$. A higher market Sharpe makes the alphas less precisely estimated.

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Market sample moments

$$\hat{\mu}_m = \bar{R}_m - \bar{r}_f = \frac{1}{T} \sum_{t=1}^T (R_{m,t} - r_{f,t})$$

$$\hat{\sigma}_m^2 = \frac{1}{T} \sum_{t=1}^T (R_{m,t} - r_{f,t} - \hat{\mu}_m)^2$$

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Residual covariance matrix

$$\hat{\Sigma} = \frac{1}{T} \sum_{t=1}^T \hat{\epsilon}_t \hat{\epsilon}_t'$$

where $\hat{\epsilon}_t = (\hat{\epsilon}_{1,t}, \dots, \hat{\epsilon}_{N,t})'$ is the $(N \times 1)$ vector of residuals at t .

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From individual t -tests to the joint F -statistic

One alpha at a time gives a t ; all alphas together give a quadratic form.

One asset at a time

Under $H_0 : \alpha_i = 0$,

$$\frac{\hat{\alpha}_i}{\text{se}(\hat{\alpha}_i)} \sim t, \quad \frac{\hat{\alpha}_i^2}{[\text{se}(\hat{\alpha}_i)]^2} \sim F.$$

Large $J_1 \Rightarrow$ joint $\hat{\alpha}$ is far from zero $\Rightarrow$ reject the asset pricing model.

Multiplying by $(T - N - 1)/N$ rescales the quadratic form to an exact F -distribution.

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All assets jointly: the quadratic form

$$J = \frac{\hat{\alpha}'\hat{\alpha}}{\text{Var}(\hat{\alpha})} = \hat{\alpha}' \left[\frac{1}{T} \left(1 + \frac{\hat{\mu}_m^2}{\hat{\sigma}_m^2} \right) \hat{\Sigma} \right]^{-1} \hat{\alpha}$$

$\sim F\text{-dist (large-sample)}.$

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GRS test statistic: exact small-sample form

$$J_1 = \frac{T - N - 1}{N} \cdot \frac{\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}}{1 + \hat{\mu}_m^2/\hat{\sigma}_m^2} \sim F_{N, T-N-1}$$

Large $J_1 \Rightarrow$ joint $\hat{\alpha}$ is far from zero $\Rightarrow$ reject the asset pricing model.

Multiplying by $(T - N - 1)/N$ rescales the quadratic form to an exact F -distribution.

The key identity: $\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}$ is a Sharpe-ratio gap

The quadratic form has a clean economic interpretation.

The fundamental identity

$$\alpha' \Sigma^{-1} \alpha = \underbrace{\frac{\mu_q^2}{\sigma_q^2}}_{\text{squared Sharpe of tangency}} - \underbrace{\frac{\mu_m^2}{\sigma_m^2}}_{\text{squared Sharpe of } M}$$

q : the in-sample tangency portfolio from the N test assets *and* the benchmark.

Economic content: J_1 measures how far the in-sample tangency Sharpe exceeds the benchmark's. Under H_0 , M is the tangency portfolio, the Sharpes are equal, and $J_1 = 0$.

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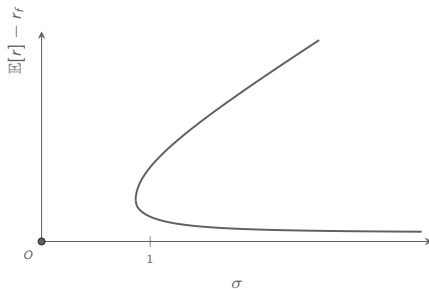
Substituting back into J_1

$$J_1 = \frac{T - N - 1}{N} \cdot \frac{\mu_q^2 / \sigma_q^2 - \mu_m^2 / \sigma_m^2}{1 + \mu_m^2 / \sigma_m^2} = \frac{T - N - 1}{N} \left[\left(\frac{\sqrt{1 + \mu_q^2 / \sigma_q^2}}{\sqrt{1 + \mu_m^2 / \sigma_m^2}} \right)^2 - 1 \right]$$

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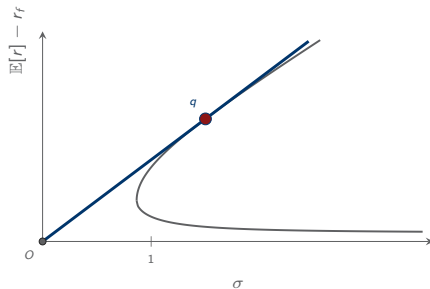
Geometric interpretation of the GRS test

Read $\overline{Od}$ vs $\overline{Oc}$ off the mean–standard-deviation plot.



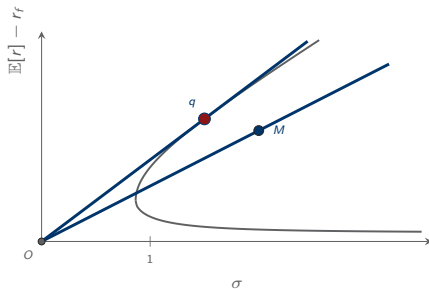
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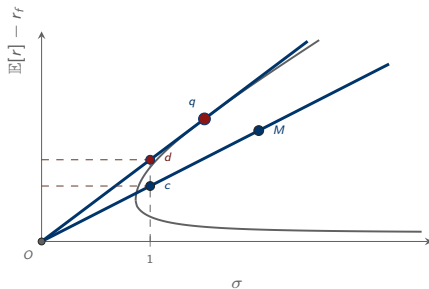
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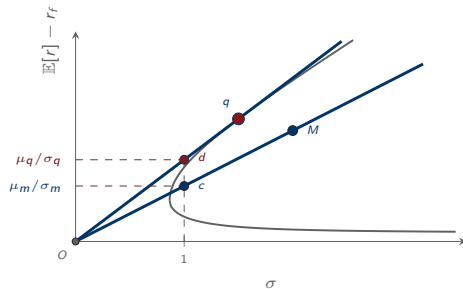
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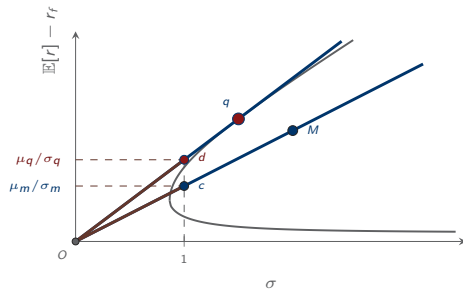
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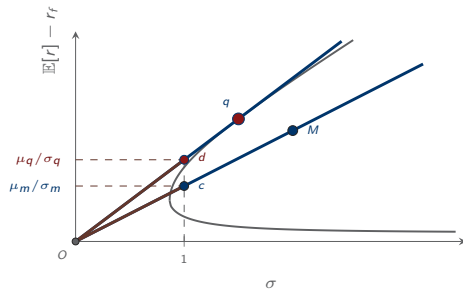
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Pythagorean reading

At $\sigma = 1$ a ray's height equals its slope (its Sharpe ratio), so each segment is the **hypotenuse** of a right triangle with base 1:

$$\overline{Od}^2 = 1^2 + \frac{\mu_q^2}{\sigma_q^2}, \quad \overline{Oc}^2 = 1^2 + \frac{\mu_m^2}{\sigma_m^2}$$

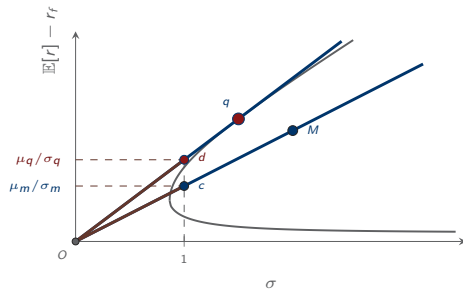
Comparing the two hypotenuses,

$$\left(\overline{Od}/\overline{Oc}\right)^2 - 1 = \frac{\mu_q^2/\sigma_q^2 - \mu_m^2/\sigma_m^2}{1 + \mu_m^2/\sigma_m^2}$$

$$\Rightarrow J_1 = \frac{T - N - 1}{N} \left[\left(\overline{Od}/\overline{Oc}\right)^2 - 1 \right]$$

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$$\overline{Od}^2 = 1^2 + \frac{\mu_q^2}{\sigma_q^2}, \quad \overline{Oc}^2 = 1^2 + \frac{\mu_m^2}{\sigma_m^2}$$

Comparing the two hypotenuses,

$$\begin{aligned} \left(\overline{Od}/\overline{Oc}\right)^2 - 1 &= \frac{\mu_q^2/\sigma_q^2 - \mu_m^2/\sigma_m^2}{1 + \mu_m^2/\sigma_m^2} \\ \Rightarrow J_1 &= \frac{T - N - 1}{N} \left[\left(\overline{Od}/\overline{Oc}\right)^2 - 1 \right] \end{aligned}$$

Equivalence

Testing the joint $\alpha = \mathbf{0}$ is **equivalent** to testing whether M is the tangency portfolio: how much further from the origin is d than c ?

The empirical evidence at a glance

Early evidence (1970s): supportive

Black, Jensen & Scholes (1972): portfolio-based tests

Fama & MacBeth (1973): two-pass cross-sectional regressions

Blume & Friend (1973)

All find evidence broadly **in favor** of the CAPM

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Anomalies emerge (late 1970s onward)

Anomalies = firm characteristics that improve the ex-post Sharpe ratio of assets above the market portfolio's, or predict average returns better than β .

Size, BE/ME, leverage, E/P, momentum, investment, profitability, . . .

"A whole host of other anomalies" — in subsequent weeks

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Why the CAPM remains useful despite rejection

A clean, parsimonious benchmark for cost of capital

A universally understood reference point for performance evaluation

Conditional, multifactor, or behavioral fixes *build on* CAPM intuition

Multifactor models: Fama–French (1993)

Fama–French (1993): five common risk factors

If size and BE/ME explain returns, what do they proxy for? Build a model.

The five factors

Three stock factors:

Mkt — r_f : excess return on a value-weighted market proxy

SMB (Small Minus Big): size factor

HML (High Minus Low): book-to-market factor

Two bond factors:

TERM: long-term gov bond return — 1-month T-bill (term-structure / interest-rate risk)

DEF: long-term corporate bond return — long-term gov bond return (default risk)

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Three extensions over FF92

1. Expand the asset set to include **bonds** as well as stocks.
2. Include **bond-market factors** from the term and default structure of interest rates.
3. Use **time-series** (not cross-sectional) tests — regression slopes are factor loadings with a clean risk-sensitivity interpretation.

Constructing SMB and HML: the 2×3 sort

Independent sorts on size and BE/ME, value-weighted.

Construction (June of each year t)

1. Sort all stocks by ME using the **NYSE median** $\Rightarrow$ Small (S), Big (B).
2. Sort by BE/ME using NYSE 30/70 breakpoints $\Rightarrow$ Low (L), Medium (M), High (H).
3. Intersect to form 6 portfolios: SL, SM, SH, BL, BM, BH.
4. Compute monthly value-weighted returns; rebalance every June.

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The factor returns

$$\text{SMB}_t = \frac{1}{3} \left(R_{SL} + R_{SM} + R_{SH} \right)_t - \frac{1}{3} \left(R_{BL} + R_{BM} + R_{BH} \right)_t$$

$$\text{HML}_t = \frac{1}{2} \left(R_{SH} + R_{BH} \right)_t - \frac{1}{2} \left(R_{SL} + R_{BL} \right)_t$$

By construction, $\text{Corr}(\text{SMB}, \text{HML}) \approx 0$.

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$$\text{SMB}_t = \frac{1}{3} (R_{SL} + R_{SM} + R_{SH})_t - \frac{1}{3} (R_{BL} + R_{BM} + R_{BH})_t$$

$$\text{HML}_t = \frac{1}{2} (R_{SH} + R_{BH})_t - \frac{1}{2} (R_{SL} + R_{BL})_t$$

By construction, $\text{Corr}(\text{SMB}, \text{HML}) \approx 0$.

Why these design choices?

NYSE breakpoints: avoid swamping small portfolios with tiny NASDAQ stocks

Value-weighted: mimicking portfolios minimize firm-specific variance and reflect tradable opportunities

Three BE/ME groups: BE/ME effect is stronger than size, so finer sort on BE/ME

Exclude negative BE, ADRs, REITs, units of beneficial interest

Why size and BE/ME?

The factors are motivated by their connection to firm fundamentals.

Fama–French (1995, JF): accounting fundamentals

High BE/ME firms have low earnings on assets, and the depressed earnings persist for at least 5 years.

Low BE/ME firms have high, persistent earnings.

Small firms suffer more in recessions and fail to recover as quickly as large firms.

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The risk-based interpretation

Size and BE/ME proxy for sensitivity to common, undiversifiable shocks to the **distress / earnings cycle** of the cross-section. Investors who care about these risks demand premia for holding small and value stocks — a multifactor extension of CAPM intuition.

The 32 dependent variables

A wide cross-section of stocks and bonds, designed to maximize spread.

25 stock portfolios

Independent 5×5 sort on size (ME) and BE/ME using NYSE quintile breakpoints

Designed to **maximize dispersion in expected returns**

Value-weighted monthly returns, July t to June $t + 1$

Rebalanced every June

Robustness: also examine sorts on E/P and D/P

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7 bond portfolios

2 government bond portfolios: 1–5 year and 6–10 year maturities

5 corporate bond portfolios by Moody's credit rating: Aaa, Aa, A, Baa, <Baa (LG)

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5 corporate bond portfolios by Moody's credit rating: Aaa, Aa, A, Baa, <Baa (LG)

Why include bonds?

A serious asset pricing model should explain stocks *and* bonds together

The flat cross-section of bond returns is informative — a successful model with low premia on TERM/DEF should still capture bond *covariation*

Table 1: summary stats on the 25 size–BE/ME portfolios

Average monthly excess returns, std deviations, t-stats; July 1963 – December 1991, 342 months.

Size \ BE/ME	Means (%)					Std deviations (%)					t-stat for mean				
	Low	2	3	4	High	Low	2	3	4	High	Low	2	3	4	High
Small	0.39	0.70	0.79	0.88	1.01	7.76	6.84	6.29	5.99	6.27	0.93	1.88	2.33	2.73	2.97
2	0.44	0.71	0.85	0.84	1.02	7.28	6.42	5.85	5.33	6.06	1.11	2.05	2.69	2.91	3.11
3	0.43	0.66	0.68	0.81	0.97	6.71	5.71	5.27	4.92	5.69	1.18	2.12	2.39	3.04	3.15
4	0.48	0.35	0.57	0.77	1.05	5.97	5.44	5.03	4.95	5.75	1.49	1.19	2.08	2.88	3.36
Big	0.40	0.36	0.32	0.56	0.59	4.95	4.70	4.38	4.27	4.85	1.50	1.42	1.34	2.43	2.26

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	Low	2	3	4	High	Low	2	3	4	High	Low	2	3	4	High
Small	0.39	0.70	0.79	0.88	1.01	7.76	6.84	6.29	5.99	6.27	0.93	1.88	2.33	2.73	2.97
2	0.44	0.71	0.85	0.84	1.02	7.28	6.42	5.85	5.33	6.06	1.11	2.05	2.69	2.91	3.11
3	0.43	0.66	0.68	0.81	0.97	6.71	5.71	5.27	4.92	5.69	1.18	2.12	2.39	3.04	3.15
4	0.48	0.35	0.57	0.77	1.05	5.97	5.44	5.03	4.95	5.75	1.49	1.19	2.08	2.88	3.36
Big	0.40	0.36	0.32	0.56	0.59	4.95	4.70	4.38	4.27	4.85	1.50	1.42	1.34	2.43	2.26

What the table says

Size effect: average returns fall with size (read down within each BE/ME column).

Value effect: average returns rise with BE/ME (read across) — value premium $\approx 0.6\%$ /mo in the extreme corner.

Standard deviations fall monotonically with size; small-growth corner is the most volatile.

Table 3: bond-market factors on stocks (m , d , t -stats)

Regress excess returns on TERM and DEF only: $R_i - r_f = a + m \text{TERM} + d \text{DEF} + e$.

Size \ BE/ME	slope m on TERM					slope d on DEF				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.93	0.90	0.89	0.86	0.89	1.39	1.31	1.33	1.45	1.52
2	0.99	0.96	0.99	1.01	0.98	1.26	1.28	1.35	1.38	1.41
3	0.99	0.94	0.94	0.95	0.99	1.21	1.19	1.25	1.24	1.21
4	0.92	0.95	0.97	1.05	1.03	0.96	1.01	1.13	1.21	1.22
Big	0.82	0.82	0.80	0.80	0.77	0.78	0.73	0.78	0.83	0.89

Size \ BE/ME	$t(m)$					$t(d)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	5.02	5.50	5.95	6.08	6.01	3.96	4.27	4.73	5.45	5.45
2	5.71	6.32	7.29	8.34	6.92	3.84	4.47	5.28	6.05	5.29
3	6.25	7.10	7.80	8.50	7.60	4.05	4.74	5.49	5.89	4.88
4	6.58	7.57	8.53	9.64	7.83	3.65	4.28	5.25	5.89	4.92
Big	7.14	7.60	8.09	8.26	6.84	3.59	3.60	4.18	4.56	4.15

Both factors highly significant for every portfolio; loadings on TERM are around 1, loadings on DEF are largest for small stocks. 39/53

Table 3: R^2 on stocks, plus bond portfolios

Size \ BE/ME	R^2 (stocks)					$s(e)$, residual s.e.				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.06	0.08	0.09	0.10	0.10	7.50	6.57	6.00	5.68	5.95
2	0.08	0.10	0.13	0.17	0.12	6.97	6.09	5.45	4.87	5.69
3	0.10	0.12	0.15	0.17	0.14	6.38	5.35	4.86	4.48	5.28
4	0.11	0.14	0.17	0.21	0.15	5.63	5.04	4.57	4.39	5.31
Big	0.13	0.15	0.16	0.17	0.12	4.61	4.33	4.00	3.89	4.55

Excess returns on government and corporate bonds

	1-5G	6-10G	Aaa	Aa	A	Baa	LG
m	0.45	0.72	1.02	0.99	1.00	1.01	0.81
$t(m)$	31.73	38.80	99.94	130.44	139.80	56.24	18.05
d	0.25	0.27	0.94	0.96	1.02	1.10	1.01
$t(d)$	9.51	7.85	48.95	67.54	75.74	32.33	11.95
R^2	0.79	0.87	0.97	0.98	0.98	0.90	0.49
$s(e)$	0.57	0.75	0.41	0.30	0.29	0.72	1.80

Bond factors leave most stock variation unexplained ($R^2 \in [0.06, 0.21]$) but price bonds excellently ($R^2 \approx 0.97$ – 0.98 top-grade).

Yet $\overline{\text{TERM}}$ and $\overline{\text{DEF}}$ are too small to explain cross-sectional variation in average stock returns.

Table 4: the market factor alone (b , $t(b)$)

One-factor regression: $R_i - r_f = a + b(R_M - r_f) + e$.

Size \ BE/ME	slope b on $Mkt - r_f$					$t(b)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.40	1.26	1.14	1.06	1.08	26.33	28.12	27.01	25.03	23.01
2	1.42	1.25	1.12	1.02	1.13	35.76	35.56	33.12	33.14	29.04
3	1.36	1.15	1.04	0.96	1.08	42.98	42.52	37.50	35.81	31.16
4	1.24	1.14	1.03	0.98	1.10	51.67	55.12	46.96	37.00	32.76
Big	1.03	0.99	0.89	0.84	0.89	51.92	61.51	43.03	35.96	27.75

Size \ BE/ME	R^2					$s(e)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.67	0.70	0.68	0.65	0.61	4.46	3.76	3.55	3.56	3.92
2	0.79	0.79	0.76	0.76	0.71	3.34	2.96	2.85	2.59	3.25
3	0.84	0.84	0.80	0.79	0.74	2.65	2.28	2.33	2.26	2.90
4	0.89	0.90	0.87	0.80	0.76	2.01	1.73	1.84	2.21	2.83
Big	0.89	0.92	0.84	0.79	0.69	1.66	1.35	1.73	1.95	2.69

All 25 stock betas highly significant ($t > 20$). R^2 tops out around 0.92, leaving real explanatory work for size and value factors. 41/53

Table 4: the market factor on bond portfolios

Dep var: excess bond returns	1–5G	6–10G	Aaa	Aa	A	Baa	LG
b	0.08	0.13	0.19	0.20	0.21	0.22	0.30
$t(b)$	5.24	5.57	7.53	8.14	8.42	8.73	11.90
R^2	0.07	0.08	0.14	0.16	0.17	0.18	0.29
$s(e)$	1.21	1.95	2.17	2.05	2.05	2.12	2.12

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b	0.08	0.13	0.19	0.20	0.21	0.22	0.30
$t(b)$	5.24	5.57	7.53	8.14	8.42	8.73	11.90
R^2	0.07	0.08	0.14	0.16	0.17	0.18	0.29
$s(e)$	1.21	1.95	2.17	2.05	2.05	2.12	2.12

What the bond row says

The market factor **captures some** bond return variation, especially for junk bonds (LG: $R^2 = 0.29$).

But b is small (0.08–0.30) compared with $R^2 = 0.07$ –0.29, so most bond variation requires the bond-market factors (TERM, DEF).

Stock and bond markets share *some* common variation through the market factor, but not most of it.

Table 5: SMB and HML alone, loadings on 25 stock portfolios

Regress on the size and value factors only: $R_i - r_f = a + s \text{SMB} + h \text{HML} + e$.

Size \ BE/ME	slope s on SMB					$t(s)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.93	1.73	1.63	1.59	1.67	22.52	21.38	21.88	22.30	22.16
2	1.52	1.46	1.35	1.18	1.40	17.23	17.68	17.08	15.47	16.42
3	1.28	1.12	1.05	0.93	1.16	14.43	13.89	13.42	12.13	13.45
4	0.86	0.82	0.77	0.72	0.95	10.16	9.64	9.29	8.57	10.02
Big	0.28	0.35	0.22	0.29	0.44	3.70	4.39	2.79	3.69	5.02

Size \ BE/ME	slope h on HML					$t(h)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.95	0.57	-0.35	-0.18	0.01	-9.72	-6.19	-4.10	-2.20	0.16
2	-1.23	-0.66	-0.38	-0.16	0.00	-12.25	-7.02	-4.20	-1.82	0.05
3	-1.09	-0.65	-0.31	-0.11	-0.01	-10.84	-7.07	-3.43	-1.23	-0.12
4	-1.11	-0.65	-0.36	-0.11	-0.01	-11.43	-6.69	-3.80	-1.12	-0.09
Big	-1.07	-0.65	-0.42	-0.06	0.08	-12.46	-7.07	-4.64	-0.66	0.81

s falls monotonically with size; h rises monotonically with BE/ME. Loadings are huge ($|t| > 10$ in many cells) – a strong signal that these factors are common.

Table 5: R^2 on stocks, plus bond portfolios

Size \ BE/ME	R^2 (stocks)					$s(e)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.65	0.60	0.60	0.60	0.59	4.57	4.31	3.98	3.79	4.01
2	0.59	0.53	0.49	0.42	0.44	4.68	4.41	4.20	4.06	4.53
3	0.51	0.43	0.37	0.31	0.35	4.71	4.31	4.19	4.10	4.60
4	0.43	0.30	0.24	0.18	0.23	4.53	4.55	4.40	4.48	5.06
Big	0.34	0.18	0.08	0.04	0.06	4.02	4.27	4.20	4.19	4.69

Excess returns on government and corporate bonds

	1-5G	6-10G	Aaa	Aa	A	Baa	LG
s	-0.02	-0.06	-0.00	0.00	0.03	0.09	0.19
$t(s)$	-0.66	-1.50	-0.15	0.22	0.77	1.99	4.19
h	0.00	-0.03	-0.02	-0.01	-0.00	0.02	0.00
$t(h)$	0.24	-0.71	-0.45	-0.22	-0.05	0.46	0.15
R^2	-0.00	0.00	-0.00	-0.00	-0.00	0.00	0.04
$s(e)$	1.26	2.03	2.34	2.24	2.25	2.34	2.46

Big-low BE/ME corner ($R^2=0.04$): SMB/HML leave big growth stocks unexplained. High-grade bonds: no loading on SMB/HML, $R^2 \approx 0$. Junk bonds (LG): $s = 0.19$, $t = 4.19$ – the size factor reaches down to high-yield credit.

Table 6: three-factor loadings, b (market) and s (size)

$$R_i - r_f = a + b(R_M - r_f) + s \text{ SMB} + h \text{ HML} + e.$$

		<i>b on Mkt-r_f</i>					<i>t(b)</i>				
Size \ BE/ME		Low	2	3	4	High	Low	2	3	4	High
Small		1.04	1.02	0.95	0.91	0.96	39.37	51.80	60.44	59.73	57.89
2		1.11	1.06	1.00	0.97	1.09	52.49	61.18	55.88	61.54	65.52
3		1.12	1.02	0.98	0.97	1.09	56.88	53.17	50.78	54.38	52.52
4		1.07	1.08	1.04	1.05	1.18	53.94	53.51	51.21	47.09	46.10
Big		0.96	1.02	0.98	0.99	1.06	60.93	56.76	46.57	53.87	38.61

		<i>s on SMB</i>					<i>t(s)</i>				
Size \ BE/ME		Low	2	3	4	High	Low	2	3	4	High
Small		1.46	1.26	1.19	1.17	1.23	37.92	44.11	52.03	52.85	50.97
2		1.00	0.98	0.88	0.73	0.89	32.73	38.79	34.03	31.66	36.78
3		0.76	0.65	0.60	0.48	0.66	26.40	23.39	21.23	18.62	21.91
4		0.37	0.33	0.29	0.24	0.41	12.73	11.11	9.81	7.38	11.01
Big		-0.17	-0.12	-0.23	-0.17	-0.05	-7.18	-4.51	-7.58	-6.27	-1.18

Market betas tight around 1; SMB loadings s decline monotonically with size (positive for small, slightly negative for big). Huge t -stats throughout.

Table 6 (continued): three-factor loadings, h (value)

Size \ BE/ME	<i>h on HML</i>					<i>t(h)</i>				
	Low	2	3	4	High	Low	2	3	4	High
Small	−0.29	0.08	0.26	0.40	0.62	−6.47	2.35	9.66	15.53	22.24
2	−0.52	0.01	0.26	0.46	0.70	−14.57	0.41	8.56	17.24	24.80
3	−0.38	0.00	0.32	0.51	0.68	−11.26	0.05	9.75	16.88	19.39
4	−0.42	0.04	0.30	0.56	0.74	−12.51	1.04	8.83	14.84	17.09
Big	−0.46	0.00	0.21	0.57	0.76	−17.03	0.09	5.80	18.34	16.24

Table 6 (continued): three-factor loadings, h (value)

Size \ BE/ME	h on HML					$t(h)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	−0.29	0.08	0.26	0.40	0.62	−6.47	2.35	9.66	15.53	22.24
2	−0.52	0.01	0.26	0.46	0.70	−14.57	0.41	8.56	17.24	24.80
3	−0.38	0.00	0.32	0.51	0.68	−11.26	0.05	9.75	16.88	19.39
4	−0.42	0.04	0.30	0.56	0.74	−12.51	1.04	8.83	14.84	17.09
Big	−0.46	0.00	0.21	0.57	0.76	−17.03	0.09	5.80	18.34	16.24

The defining pattern of FF93

HML loadings rise monotonically with BE/ME — **growth stocks load negatively**, value stocks load positively.

Pattern holds across *all* size buckets — the value factor is not specific to small or large.

$|t(h)|$ values up to ≈ 25 confirm HML captures genuine common variation, not noise.

Table 6: R^2 on stocks, plus bonds

Size \ BE/ME	R^2 (stocks)					$s(e)$				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.94	0.96	0.97	0.97	0.96	1.94	1.44	1.16	1.12	1.22
2	0.95	0.96	0.95	0.95	0.96	1.55	1.27	1.31	1.16	1.23
3	0.95	0.94	0.93	0.93	0.93	1.45	1.41	1.43	1.32	1.52
4	0.94	0.93	0.91	0.89	0.89	1.46	1.48	1.49	1.63	1.88
Big	0.94	0.92	0.88	0.90	0.83	1.16	1.32	1.55	1.36	2.02

Excess returns on government and corporate bonds

	1-5G	6-10G	Aaa	Aa	A	Baa	LG
b	0.10	0.18	0.25	0.25	0.26	0.27	0.34
$t(b)$	6.45	6.75	8.60	9.30	9.46	9.58	12.22
s	-0.06	-0.14	0.12	-0.11	-0.09	-0.04	0.04
$t(s)$	-2.70	-3.65	-2.89	-2.72	-2.18	-0.91	0.89
h	0.07	0.08	0.14	0.15	0.16	0.20	0.23
$t(h)$	2.66	1.83	2.77	3.26	3.51	4.08	4.75
R^2	0.10	0.12	0.17	0.20	0.20	0.22	0.33
$s(e)$	1.19	1.91	2.13	2.00	2.01	2.08	2.06

Stock $R^2 \in [0.83, 0.97]$, the highest yet. Market β s tighten around 1; SMB and HML loadings line up monotonically with the sort. Bonds still load weakly on stock factors — need TERM and DEF to fully explain bonds.

Table 9a: intercepts on 25 stock portfolios, five specs

Under each asset-pricing model, every intercept should be zero.

Size \ BE/ME	intercept a (%/month)					$t(a)$				
	Low	2	3	4	High	Low	2	3	4	High
<i>(i) $R_i - r_f = a + m \text{ TERM} + d \text{ DEF} + e$ (bond factors only)</i>										
Small	0.31	0.62	0.71	0.80	0.92	0.75	1.73	2.20	2.61	2.87
2	0.35	0.63	0.77	0.75	0.93	0.93	1.91	2.60	2.85	3.03
3	0.34	0.58	0.60	0.73	0.89	1.00	1.99	2.28	3.01	3.11
4	0.41	0.27	0.49	0.69	0.96	1.34	1.01	1.96	2.88	3.35
Big	0.34	0.30	0.25	0.50	0.53	1.35	1.27	1.17	2.36	2.14
<i>(ii) $R_i - r_f = a + b(R_M - r_f) + e$ (CAPM)</i>										
Small	-0.22	0.15	0.30	0.42	0.54	-0.90	0.73	1.54	2.19	2.53
2	-0.18	0.17	0.36	0.39	0.53	-1.00	1.05	2.35	2.79	3.01
3	-0.16	0.15	0.23	0.39	0.50	-1.12	1.25	1.82	3.20	3.19
4	-0.05	-0.14	0.12	0.35	0.57	-0.50	-1.50	1.20	2.91	3.71
Big	-0.04	-0.07	-0.07	0.20	0.21	-0.49	-0.95	-0.70	1.89	1.41
<i>(iii) $R_i - r_f = a + s \text{ SMB} + h \text{ HML} + e$ (no market)</i>										
Small	0.24	0.46	0.49	0.53	0.55	0.97	1.92	2.24	2.52	2.49
2	0.52	0.58	0.64	0.58	0.64	2.00	2.40	2.76	2.61	2.56
3	0.52	0.61	0.52	0.60	0.66	2.00	2.58	2.25	2.66	2.61
4	0.69	0.39	0.50	0.62	0.79	2.78	1.55	2.07	2.51	2.85
Big	0.76	0.52	0.43	0.51	0.44	3.41	2.23	1.84	2.20	1.70

CAPM (ii) leaves big positive intercepts in the high-BE/ME column; bond-factor-only (i) leaves them everywhere; size+value only (iii) leaves big intercepts in the big-stock corner.

Table 9a (continued): 3-factor and 5-factor intercepts

Size \ BE/ME	intercept a (%/month)					$t(a)$				
	Low	2	3	4	High	Low	2	3	4	High
<i>(iv) $R_i - r_f = a + b(R_M - r_f) + s \text{SMB} + h \text{HML} + e$ (three-factor)</i>										
Small	-0.34	-0.12	-0.05	0.01	0.00	-3.16	-1.47	-0.73	0.22	0.14
2	-0.11	-0.01	0.08	0.03	0.02	-1.24	-0.20	1.04	0.51	0.34
3	-0.11	0.04	-0.04	0.05	0.05	-1.42	0.47	-0.47	0.71	0.56
4	0.09	-0.22	-0.08	0.03	0.13	1.07	-2.65	-0.99	0.33	1.24
Big	0.21	-0.05	-0.13	-0.05	-0.16	3.27	-0.67	-1.46	-0.69	-1.41
<i>(v) Five-factor: $b \text{Mkt} + s \text{SMB} + h \text{HML} + m \text{TERM} + d \text{DEF} + e$</i>										
Small	-0.35	-0.13	-0.05	0.01	0.00	-3.24	-1.58	-0.79	0.20	0.09
2	-0.11	-0.02	0.08	0.04	0.02	-1.29	-0.24	1.10	0.67	0.29
3	-0.12	0.04	-0.03	0.06	0.05	-1.45	0.48	-0.42	0.79	0.56
4	0.08	-0.22	-0.08	0.04	0.13	1.04	-2.67	-0.94	0.47	1.23
Big	0.21	-0.05	-0.13	-0.06	-0.17	3.29	-0.72	-1.46	-0.73	-1.51

Adding *TERM* and *DEF* (spec *v*) leaves stock intercepts essentially unchanged from the three-factor model (*iv*). The bond factors are absorbed by *SMB* and *HML* in the joint regression.

Table 9c: joint GRS F -test on the intercepts

For each model, test the joint hypothesis $\alpha_1 = \dots = \alpha_{25} = 0$.

GRS F -statistics and bootstrap p -values

	(i) bond only	(ii) CAPM	(iii) SMB+HML	(iv) 3-factor	(v) 5-factor
GRS F -statistic	2.09	1.91	1.78	1.56	1.66
Bootstrap p -value	0.998	0.996	0.985	0.951	0.971

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Bottom line

Every model is rejected at conventional levels, but the three-factor model (iv) has the **smallest** F -statistic.

The bulk of the misfit comes from the **small-low-BE/ME** corner ($\alpha = -0.34\%/mo$, $t = -3.16$) — the “small growth” anomaly.

Outside that corner, intercepts are economically tiny ($|\alpha| \lesssim 0.20\%/mo$): the three-factor model performs very well in practice.

Robustness battery

Test predictability of three-factor residuals using known predictors (D/P , r_f , term spread)

January seasonals: examine month-by-month

Split-sample tests: form factors with one set of stocks, test on a different set

Portfolios sorted on E/P and D/P

Robustness checks and conclusions

Robustness battery

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Portfolios sorted on E/P and D/P

Five-factor conclusions

The **market factor** is common to all stocks and explains the stock–bond return premium.

Bond-market factors (TERM, DEF) capture substantial time-series variation in stock returns related to business-cycle conditions, but their premia are too small to explain cross-sectional variation in average returns.

Open question: *why* are size and BE/ME related to average returns? Risk? Mispricing? Data mining (Ball, 1978)?

Takeaways

What rejecting the CAPM buys us

The empirical record

CAPM works *descriptively* — average returns line up with co-movement in good times vs bad

CAPM fails *quantitatively* — β alone does not capture the cross-section

Size and BE/ME (and later: momentum, profitability, investment) are robust predictors

The three-factor model (Mkt, SMB, HML) absorbs most of these effects

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Three responses to rejection

1. **Multifactor models:** FF 3-factor, Carhart 4-factor, FF 5-factor — risks beyond the market
2. **The tests are flawed:** Roll's critique, conditional information, sample selection, measurement
3. **Behavioral:** mispricing, sentiment, limits to arbitrage — no risk model fits some phenomena

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Why the CAPM still matters

A useful benchmark, a teaching device, the foundation of every multifactor extension — and the reference point for thinking about systematic vs idiosyncratic risk.

Cross-sectional and time-series tests: summary

$$r_{it} - r_{ft} = \alpha_i + \beta_i(R_{Mt} - r_{ft}) + \varepsilon_{it}$$

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Cross-sectional tests

Fama–MacBeth (1973): month-by-month cross-section, average the slopes

Test: $\mathbb{E}[\gamma_2] = \mathbb{E}[\tilde{R}_M] - r_f > 0$

Issues: errors-in-variables, microstructure, regression phenomenon

Fama–French (1992): β is dead; size and BE/ME live

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The big picture

Asset pricing tests are ultimately tests about whether *some* portfolio is mean–variance efficient — the only question is which one, and which factors define it.